

What the Survey Missed

Teacher and presenter guide - bounded synthetic occupancy investigation - Murray State University

Purpose and learning target

Students distinguish observed fraction ever detected, realized occupied-site fraction in one synthetic scenario, occupancy probability (ψ), and detection probability (p). Occupancy means whether a site is used in this model; it is not abundance or a number of animals.

Target claim: fewer detections do not by themselves establish lower occupancy. Detection probability can differ between habitats.

Every round visits the same 24 Habitat A and 24 Habitat B sites: 48 site-visits. A site click inspects a fixed detection history; it never changes the sample.

Current app flow

- Run fixed round 1 across all 48 sites.
- Inspect observed fraction ever detected and choose the cautious claim: a difference can reflect occupancy, detection probability, or both.
- Only then choose fixed rounds 2 and 3, transfer, optional likelihood, or reveal.
- After reveal, the same arrays distinguish detections, occupied-but-missed sites, and unoccupied sites. Further rounds are exploration, not a blind new test.

Controls: ? reopens how-to. **Settings** contains presentation mode, openable presenter notes, reset responses, practice replay of the same seed, and a new seeded scenario.

Declared synthetic model

Habitat	Generating occupancy ψ	Detection p	Meaning
A	0.8	0.25	Higher generating occupancy probability, lower detection probability.
B	0.5	0.8	Lower generating occupancy probability, higher detection probability.

Sites remain occupied or unoccupied across the three visits (closure). Given occupancy, detections are conditionally independent; there are no false positives; and p is constant within a habitat. These are teaching assumptions, not claims about local wildlife.

Why raw detections can mislead

For K visits, expected fraction ever detected is $\psi \times [1 - (1 - p)^K]$.

Visits K	A expected ever detected	B expected ever detected	Implication
1	0.20	0.40	B is expected to have more detections although A has greater ψ .
3	0.4625	0.496	More visits improve evidence but do not guarantee a finite ranking matches occupancy.

Do not score the ranking. After reveal, compare observed fraction, realized occupancy, generating ψ , and p by name. A seeded realization can differ from either expectation.

Optional likelihood view

The app uses observed K -visit histories for one habitat only. For an ordered history with d detections, likelihood is $\psi^d (1-\psi)^{(K-d)}$ when d is positive; for an all-zero history it is $(1-\psi)^K$.

Interpret carefully: shading is relative likelihood on a bounded ψ/p grid. It is not posterior probability or a calibrated confidence interval. Multiply exact ordered-history probabilities over sites; do not add a binomial coefficient to one particular history.

With one visit, detections chiefly identify ψ/p . A high- ψ /low- p and lower- ψ /higher- p combination can fit the same observed fraction. The app supplies axis labels plus keyboard/touch readout rather than a unique occupancy estimate.

Facilitation and boundaries

- Before round 1, contrast “not detected” with “not present.”
- After round 1, ask for the unit of effort: 48 site-visits, not 48 animals counted.
- Before another round, ask what changes (histories) and what stays fixed (sites).
- For transfer, ask whether a different detection process could explain fewer detections.

This activity teaches detection confounding. It does not estimate a real species, habitat, population, conservation priority, or field-survey result.

Sources and attribution

MacKenzie, D. I., et al. (2002), “Estimating Site Occupancy Rates When Detection Probabilities Are Less Than One,” *Ecology*, 83(8), 2248-2255. U.S. Geological Survey: Estimating site occupancy rates when detection probabilities are less than one.

U.S. Geological Survey, RPresence occupancy workshop, 2023 online, for occupancy/detection-history context. These are source context, not course assignment information or endorsement.